

Progress Update: Model Structure and Validation Cases

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1 Overview

In this document the basic structure of a trajectory optimization-based control planning method and example use cases are presented. The document covers the basic structure of the problem, the dynamics model that form the basis of the tool, as well as the aerodynamic and gust models which are being used for testing. The intent is to demonstrate that the solution tools that have been constructed provide heuristically correct responses when posed with complex dynamic questions.

The basic problem we will be asking is for the optimizer to find a flyable trajectory for an aircraft given some initial and final conditions, the aircraft dynamics, and other constraints.

2 Simulation Details

This section details the construction of the optimal control problem and the detail of the dynamics model of the aircraft. Sections 2.1, 2.2 relate to the general construction of the optimal control problem and dynamics model. Sections 2.3, 2.4 relate to the aerodynamic and gust models as currently constructed for the validation cases.

2.1 Problem Formulation

The problem statement shown in eq. (1) sets up the trajectory planning question as a *direct transcription* optimization problem over the trajectory $\mathbf{x}$ and $\mathbf{u}$ as described by Tedrake [2]. This sets up a problem which involves searching for a trajectory $\mathbf{x}, \mathbf{u}$ that optimizes the function $\ell + J$ and which also satisfies the constraints which include the system dynamics as well as others. In the present simulation structure the problem is parameterized in time through the discrete index k and discrete time index $k\Delta t$ which is provided from $[1, N]$, where N is the length of the simulation.

$$\begin{aligned} \min_{\mathbf{x}[k], \mathbf{u}[k]} \quad & \ell(\mathbf{x}[N], \mathbf{u}[N]) + \sum_{k=0}^{k=N} J(\mathbf{x}[k], \mathbf{u}[k]) \\ \text{s.t.} \quad & \mathbf{x}[k+1] = f(\mathbf{x}[k], \mathbf{u}[k]) \\ & g(\mathbf{x}[k], \mathbf{u}[k]) \leq 0 \\ & h(\mathbf{x}[k], \mathbf{u}[k]) = 0 \\ & + \text{other constraints...} \end{aligned} \tag{1}$$

The constraints on the problem consist of the system dynamics $f(\mathbf{x}, \mathbf{u})$, and equality and inequality constraints on $h(\mathbf{x}, \mathbf{u})$, $g(\mathbf{x}, \mathbf{u})$. These may be used for a variety of purposes such as setting the

ground plane (altitude ≥ 0), or rate limits on controls like the elevator. They may also be used to bound $f(\mathbf{x}, \mathbf{u})$ such as limiting rotation rates, velocity, etc. with the proper setup. This method is mostly intended as a trajectory planning technique rather than an in-the-loop controller technique; however, according to [2] if the problem is constructed to be quadratic and convex these can be used online as controllers for manipulation and other under-actuated problems.

2.2 Aircraft Motion Modelling

For the present simulation only the longitudinal aircraft dynamics are modelled. The aircraft is assumed to be perfectly rigid and is represented as a point mass located at the mass centroid of the aircraft (which in this case is arbitrary). The point mass has freedom to move horizontally and vertically, and rotate about the pitch axis. The state and state derivatives are given in eqs. (2) and (3).

$$\mathbf{x} = \{x^e, z^e, u^b, w^b, \theta, q\} \quad (2)$$

$$\dot{\mathbf{x}} = \{u^e, w^e, a_x^b, a_z^b, q, \dot{q}\} \quad (3)$$

The body and earth axes systems used for the motion model are taken from [1]. Conversion between the two axes system is given by eq. (4).

$$T_e^b = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \quad (4)$$

Forces from all sources are summed in the body axes system; The m and I_{yy} values of the model are then used to get accelerations in the body axes as shown in eqs. (5) to (7). To compute the motion in the earth frame the velocities in the body frame are converted using 4.

$$a_x = \frac{1}{m} \sum F_{x,b} = F_{x,\text{wing}} + F_{x,\text{tail}} + F_{x,\text{gravity}} + F_{x,\text{propulsor}} + \dots \quad (5)$$

$$a_z = \frac{1}{m} \sum F_{z,b} = F_{z,\text{wing}} + F_{z,\text{tail}} + F_{z,\text{gravity}} + F_{z,\text{propulsor}} + \dots \quad (6)$$

$$\dot{q} = \frac{1}{I_{yy}} \sum M_{y,b} = M_{y,\text{wing}} + M_{y,\text{tail}} + M_{y,\text{propulsor}} + \dots \quad (7)$$

Figure 1: Point mass representation of the aircraft system showing relationship between axes systems

Forces and moments generated by $f(\mathbf{x}, \mathbf{u})$ are covered in section 2.3 and are model dependent. Forces are always computed in the most 'natural' axis system then converted back to the body axis for summation and computation of the accelerations

Figure 2: The basic wing-tail-propulsor aircraft model used by the simulation

2.3 Aerodynamics Model

The current simulation uses a simple thin-airfoil theory model of a plane consisting of a wing, stabilizer, and propulsor shown in fig. 2. $\mathbf{u} = \{\delta_f, \delta_e, \delta_T\}$. The influence of the flap and elevator are determined using the flap modification of thin airfoil theory and the propulsor is a simple proportional model with maximum thrust defined off a fraction of the aircraft mass.

- full model definition
- aerodynamics / moments summation
- how coefficients are calculated
- how drag coefficients are calculated
- throttle forces

Forces from the wing tail and propulsor are calculated independently and summed at the point mass using the model below. The pitching moment from the tail comes from both the pitching moment from the tail as well as the offset from the tail length from the quarter chord. The center of mass and quarter chord are assumed to be coincident.

$$L_{\text{tot}} = L_{\text{wing}} + \quad (8)$$

$$M_{\text{tot}} \quad (9)$$

$$D_{\text{tot}} \quad (10)$$

These are rotated into the body frame via eq. (11) and all of the forces summed in the body frame via equation eqs. (5) to (7).

$$T_a^b = \begin{bmatrix} \cos(\alpha) & \sin(\alpha) \\ -\sin(\alpha) & \cos(\alpha) \end{bmatrix} \quad (11)$$

- two airfoil element model with propulsor underslung
- aerodynamics are thin-airfoil-theory aerodynamics
- all quasi steady
- drag is const + quadratic term

2.4 Gust Model

- intent right now is only to model thermal updraft and down draft
- gusts are implemented as fields of vertical velocity that changes only along x
- show general shape of gust with intense core and much softer edges
- gusts are defined by a core x location, multiple gusts can be overlayed onto the same plane

Figure 3: comparison of cruise response to vertical gust

- typical gust ranges

Gusts are modelled as atmospheric velocity fields $G(x^e, z^e)$ which alter the local x,z velocity components with the atmospheric velocity $\vec{V}_G = u_{\text{atm}}\hat{i} + w_{\text{atm}}\hat{k}$

$$V_\infty = (u^b - u^{\text{atm}}) + (w^b - w^{\text{atm}}) \quad (12)$$

$$\alpha = \arctan\left(\frac{w^b - w^{\text{atm}}}{u^b - u^{\text{atm}}}\right) \quad (13)$$

3 Results

The models from the previous sections are implemented using table 1

3.1 Aircraft Model Specifications

Table 1: Table of aircraft model specifications

mass	10 kg	y-inertia	3 N/m
span	3.x m	chord	0.6 m
AR	5	tail span	1
tail chord	0.3	tail AR	3

3.2 Cruise

compare steady state to single vertical gust
constraints and start / end conditions

$$J(x[k], u[k]) = \sum_{k=1}^N (z_e[k] - z_e[0])^2 \quad (14)$$

We would expect with an updraft the the aircraft will pitch down into the updraft to keep altitude the same then ease off as it passes the core of th gust. Results show just that. Sim does allow for (extraordinarily) large deviations from altitude in the vent control insufficient for thermal

3.3 Landing

$$J(x[k], u[k]) = \sum_{k=1}^N (z_e[k] - z_e[0])^2 \quad (15)$$

4 Ongoing Work

- physics rather than proportional-based propulsor model
- blowing term reintroduction (calculate things like $C_{T'}$, ΔC_J) to then use blowing CL-CX models
- multi-element wing model similar to MDs flight dynamics sim paper
- problem reparameterization into either track length or x-position, testing
- start work on 6DOF implementation

References

- [1] Mark Drela. *Flight vehicle aerodynamics*. MIT press, Cambridge Mass., 2014.
- [2] Russ Tedrake. *Underactuated Robotics Algorithms for Walking, Running, Swimming, Flying, and Manipulation*.